

Experimental Data Management Guidelines

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1 Purpose

Experimental data are an important research asset and must be organized, documented, and preserved appropriately. Data collected during an experiment may be needed months or years later for further analysis, preparing publications, responding to reviewers, or extending previous research.

The purpose of these guidelines is to establish a consistent and straightforward approach to managing experimental data generated in the Control, Estimation, and Learning Laboratory (CELL).

The guiding principle is that **another researcher should be able to understand the experimental data without having to consult the person who originally performed the experiment.**

These guidelines apply to all laboratory experiments that generate substantial amounts of data.

2 Data Storage

All experiments that generate substantial amounts of data must have their data uploaded to the laboratory's shared Box folder.

- The shared Box folder serves as the central repository for experimental data generated in CELL.
- Students must not rely exclusively on local computers, personal drives, or external storage devices to preserve experimental data.
- Experimental data should be uploaded to Box promptly after data collection.
- If you do not have access to the appropriate shared Box folder, ask Dr. Goel to set one up for you.
- Original experimental measurements must be preserved. Processing or analysis should not overwrite raw data.

3 Folder Organization

Each experimental campaign should have a dedicated folder using the following naming convention:

`YYYY_MM_DD_Purpose_Initials`

For example, `2026_10_08_OpenLoopTests_AG`.

The folder name should identify the date, purpose of the experiment, and initials of the person conducting it.

The required folder structure is shown below.

```
YYYY_MM_DD_<exp_desc>_<Initials>/
|
|-- raw/                          # Original experimental measurements
|   |-- rec_1_01.mat
|   |-- rec_1_02.mat
|   |-- rec_2_01.mat
|
|-- analysis/                     # Analysis scripts and generated figures
|   |-- plot_sample.m
|   |-- sample_plot.png
|
|-- README.md                    # Experiment description and data documentation
```

4 README Documentation

Every experimental dataset must include a README.md file in the main experiment folder. The README should provide sufficient information for another researcher to understand the purpose of the experiment and interpret the generated data.

A .md file is a Markdown file, which uses a lightweight markup language similar to LaTeX but with much simpler syntax. Markdown files can be created and edited using any text editor and are automatically rendered by GitHub.

The following template should be adapted to each experiment.

```
# Experimental Details

- Date: <YYYY-MM-DD>
- Experiment name: <Name>
- Experiment purpose: <Brief description>
- Conducted by: <Name(s)>

# Purpose

<Describe what the experiment is doing and why it is being conducted.>

# Data Generated

<Describe the data generated by the experiment, including the measurements, file contents
, and the file naming convention.>

# Additional Notes

<Include any relevant observations, issues, limitations, or other information.>
```

The README should be updated as necessary to reflect the experiments performed and the data collected.

5 Data Visualization

Each experimental dataset must include a simple MATLAB or Python script in the `analysis/` folder that loads at least one sample data file and generates a representative plot.

The script should demonstrate how to access and interpret the recorded measurements. The generated figure may also be saved in the `analysis/` folder.

The goal is to ensure that another researcher can quickly load and visualize the experimental data without needing additional instructions.

6 Minimum Requirements

To avoid unnecessary administrative overhead, CELL does not require elaborate filenames or extensive documentation for every individual measurement.

However, every experimental dataset must satisfy the following minimum requirements:

1. **Organized storage:** Use a dedicated experiment folder with a descriptive name.
2. **Shared access:** Upload the experimental data to the laboratory's shared Box folder. If access is unavailable, contact Dr. Goel.
3. **Documentation:** Include a `README.md` explaining the experiment, its measurements, and the organization of the data.
4. **Data visualization:** Include a MATLAB or Python script that loads and plots at least one sample data file.
5. **Raw data preservation:** Do not overwrite or modify the original experimental measurements.

The goal is not to create additional paperwork, but to ensure that experimental data remain accessible, understandable, and useful to the laboratory long after the original experiments are completed.